

Web Scraping Project Using Python

Introduction

This project demonstrates how to use Python to scrape product data from Amazon. We will extract product names, prices, and ratings using libraries such as requests, BeautifulSoup, and pandas.

Tools and Libraries Used

1. requests to send HTTP requests to the website.
2. BeautifulSoup to parse and navigate the HTML content.
3. pandas to store and manipulate data, and save it in a CSV file.

Step 1: Import Libraries

We start by importing all the required libraries: requests, BeautifulSoup, and pandas.

Step 2: Send Request to Amazon

Define the search term (e.g., 'headphones') and set up browser-like headers to avoid being blocked by Amazon. Then send a GET request using requests.get() and parse the response using BeautifulSoup.

Step 3: Extract Product Information

Loop through the product containers in the HTML response. For each product, extract the product name, price, and rating. Store this information in lists.

Step 4: Save to CSV

Convert the collected lists into a pandas DataFrame and save the DataFrame to a CSV file called 'amazon_products.csv'.

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Enhancements

You can improve this project by:

- Scraping multiple pages.

- Using Selenium for JavaScript-loaded content.

- Scheduling scraping tasks.

- Adding a user interface using tkinter.

Conclusion

This project shows how Python can automate data collection from e-commerce sites. Web scraping is useful for market research, price tracking, and competitive analysis. However, always respect website policies and use scraping ethically.